

MyTix User Manual

1. Requirements

1.1 Prerequisites

- MySQL 8 is installed and running locally
- Java Development Kit 17 or newer
- Bash for `run.sh`
- The supplied MySQL Connector/J and OpenNLP JAR files in `lib/`

1.2 Configure the database connection

Copy `config.properties.example` to `config.properties` and enter the database URL, user, and password for your MySQL instance. In the project archive, we also provide a `config.properties` with the default username 'root' and an empty password.

The terminal shows `Database: CONNECTED` when startup succeeds after running `./run.sh`. If it shows `OFFLINE`, use main-menu option 10, check the displayed target and last error, then enter `R` to reconnect.

1.3 Initialize and run

Set up the database (run once)

1. Open a terminal and log into MySQL: `sudo mysql`
2. Create the database for the application:

```
CREATE DATABASE IF NOT EXISTS <YOUR_DB_NAME>;  
EXIT;
```

Use a clean database and follow this exact order:

1. Execute `sudo mysql <YOUR_DB_NAME> < sql/schema.sql`
2. Execute `sudo mysql <YOUR_DB_NAME> < sql/load.sql`
3. From the project root, run `./run.sh`.

If you need a clean restart, run `sql/drop.sql`, then repeat the three steps above.

`schema.sql` creates the tables and constraints. `load.sql` creates the complete sample dataset. `run.sh` compiles the Java source and starts the terminal. No manual insert or code edit is required.

At the main menu, type the number of an area and press Enter. Type `0` to go back or exit. After a result, press Enter to continue.

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Before using each option, check the IDs above the table.

Note:

All the other data information required for operations can be found using the submenu options which lets you discover valid data. For example, to book reserved seats (Main Option 4), you need a row letter and seat number; you can locate available seats by entering a performance ID (e.g., 6001) in Submenu Option 1. Similarly, before listing a resale, use Option 5 → 1 to view a customer's current tickets or Option 5 → 2 to list active resale listings for a performance.

2.1 Main option 1 – User profiles

Use the **customer IDs** ranging from **2001-2100** to test the operation.

Sub-menu	Use	Required input	Result
1	Create customer profile	name, address, unique email, birth date, card number, card-holder name, expiry date, billing postal/ZIP	New customer ID
2	Create organizer profile	name, address, unique email, birth date	New organizer ID
3	View customer profile	customer ID (e.g., 2001)	Profile, account status, masked card, expiry, billing postal/ZIP
4	Deactivate user profile	active user ID, then exact confirmation DEACTIVATE	Personal fields are anonymized; sales and ownership history remain
5	View customer order and ticket history	customer ID (e.g., 2001)	Orders, tickets, venue, price, current/ended ownership, cancellation, and refund

2.2 Main option 2 – Organizer events and performances

Use the following IDs ranges to test the operations:

1. **Organizer ID: 1001-1005**, for submenu option 1,4
2. **Genre ID: 4101-4106**, for submenu option 1
3. **Artist/team ID: 4201-4215**, for submenu option 1
4. **Event ID: 5001-5020**, for submenu option 2,3
5. **Venue ID: 3001-3008**, for submenu option 2

Sub-menu	Use	Required input	Result
1	Create event with billing	organizer ID, genre ID, title, optional description, resale-cap multiplier (default 1.20), artist/team count, each artist/team ID and unique billing rank	New event ID
2	Add performance	event ID, venue ID, YYYY-MM-DD HH:mm	New performance ID
3	Update event resale cap	event ID, multiplier at least 1.00	New cap used for future listings
4	View managed events and sales	organizer ID (e.g., 1001)	Event/performance status, original sales, refunds, resales, and transaction details

2.3 Main option 3 – Performance pricing and inventory

Use the **performance IDs** ranging from **6001-6070** to test the operations, some operations require specific valid IDs:

1. Submenu 3: **6006, 6013** — scheduled performances with no ticket-sale history
2. Submenu 4,5,6: **6001**, 6002, 6006, 6013-6018, 6020, 6023, 6026, 6029, 6032, 6035, 6038, 6040, 6042, 6044, 6046, 6048, 6050, 6052, 6054, 6056 — scheduled, pending performances

Sub-menu	Use	Required input	Result
1	View reserved inventory	performance ID (e.g., 6001)	Section, row, seat, tier, price, blocked flag, and state
2	View general-admission inventory	performance ID (e.g., 6001)	Section, tier, price, total, remaining, and sold
3	Configure all tiers and sections	eligible performance ID (6006, 6013), tier count (≥ 2), unique tier codes&prices, one tier for every displayed section	Complete replacement summary and section map
4	Update one tier price	scheduled performance ID, tier code, positive price	Updated future selling price

Sub-menu	Use	Required input	Result
5	Block reserved seat	performance ID, row, seat number (which can be found using the submenu option 1)	Seat becomes unavailable
6	Unblock reserved seat	performance ID, row, seat number (which can be found using the submenu option 1)	Seat becomes available if not sold

A sold seat cannot be blocked and cannot be made available by unblocking it.

2.4 Main option 4 – Ticket booking and cancellation

Use the scheduled, pending **performance IDs: 6001, 6002, 6006, 6013-6018, 6020, 6023, 6026, 6029, 6032, 6035, 6038, 6040, 6042, 6044, 6046, 6048, 6050, 6052, 6054, 6056**

Use the **customer IDs** ranging from **2001-2100**, excluding scalpers 2001, 2002 after R4 is run

Use the active, seven days away from performance **ticket IDs: 8001, 8003, 8004, 8007-8015, ..., 8308 - 8348**, and more...

Sub-menu	Use	Required input	Result
1	View reserved seats	performance ID (e.g., 6001)	Displays one or more available seats, including the section, row, seat number, tier, and price.
2	Book reserved seats	active, unrestricted customer ID, performance ID, seats as Row Seat# separated by commas (e.g., A 1, A 2 which can be found using the submenu option 1)	Prints a new transaction ID, new ticket IDs, and the total.
3	Book general admission	active unrestricted customer ID, performance ID, one of the displayed GA section names, positive quantity	Checks and locks capacity. A request larger than remaining capacity returns CONFLICT.
4	Cancel customer tickets	one or more ticket IDs separated by commas, optional reason. (Ticket 8004 is a success example, while ticket 8006 is a rejection example.)	Closes ownership, withdraws active listings, records cancellation, and refunds acquisition price.

Sub-menu	Use	Required input	Result
5	Cancel a performance	valid performance ID, optional reason	Marks performance cancelled, cancels all active tickets, and creates refunds in one transaction.

2.5 Main option 5 – Ticket resale and histories

Use the **customer IDs** ranging from **2001-2100**, note 2001 and 2002 will be identified as scalpers after R4 is run, and they'll be restricted to buy any ticket in submenu option 5

Use the **performance IDs** that have available resale tickets: **6001, 6002**

Use the **ticket IDs** ranging from **8001-8960** to view ticket ownership history (submenu option 6)

Sub-menu	Use	Required input	Result
1	View currently owned tickets	customer ID (e.g. 2003)	Active ticket IDs, performance IDs, and status; use this to find candidates
2	View available resale tickets	performance ID (6001, 6002)	Ticket ID, seller customer ID, and listing price
3	Resale a ticket	customer ID, currently owned active ticket ID (which can be found in submenu option 1) , listing price	Listing ID, price, and cap
4	Withdraw listing	customer ID, actively listed ticket ID (which can be found in submenu option 1)	Listing becomes withdrawn
5	Purchase resale ticket	unrestricted customer ID, actively listed ticket ID (which can be found in submenu option 2, or use 8001, 8002, 8003, 8005)	Resale transaction and listing IDs, ticket ID, purchase price
6	View ticket ownership history	ticket ID (e.g. 8007)	Every owner, acquisition transaction/price, listing, and ownership interval

2.6 Main option 6 – Attendance reviews

Use **customer IDs**:

1. who have not submitted a review: **2003**, 2004, 2011-2095
2. who have submitted a review: **2001**, 2002, 2009-2011, 2013-2015, 2021-2023, 2029-2031, 2039-2041, 2053, 2054, 2060, 2065-2067, 2078-2080, 2093-2095

Submenu	Use	Required input	Result
1	Submit review	customer ID, one displayed eligible performance ID, event rating 1-5, venue rating 1-5, nonblank comment	One review submitted from that customer
2	View submitted reviews	customer ID (e.g., 2001)	Event, venue, ratings, date, and comment

Note that customers can only submit reviews for attended/completed performance

2.7 Main option 7 –Q1-Q7

Q	Purpose and example input	Output and expected behavior
Q1	Latitude 43.6, longitude -79.3, radius 100 km (or blank for default); sort by distance, lowest price ascending, or descending	Upcoming performances inside the radius, with venue, distance, lowest price.
Q2	Postal code M5J 2X2	Upcoming performances whose venue matches the same and adjacent postal codes (M5J)
Q3	Exact address 40 Bay Street	Upcoming performances for every venue at that normalized exact address
Q4	Choose a location mode, use the coordinate, postal code, or address from Q1, Q2, Q3 example, and the time range Start date: 2026-08-08 End date: 2026-12-06 Minimum available tickets: 2	Only upcoming performances in the location/date range with at least that many total available tickets
Q5	Example: (input field can be empty) City: Toronto Segment: Music Genre: Rock Start date: 2026-08-08 End date: 2027-02-03	Performances satisfying every supplied filter; blank optional filters are ignored; price/availability apply to reserved inventory in this example

Q	Purpose and example input	Output and expected behavior
	Minimum price: 0 Maximum price: 500 Minimum available tickets: 1 Section type: reserved	
Q6	Performance 6001-6070	One row per section/tier with price, available or remaining count, sold count, and blocked reserved count
Q7	Performance 6001-6070 Example: Performance 6001, group size 4, no maximum budget	Best available tier and seats/capacity; result states whether reserved seats are consecutive and may use a nonconsecutive fallback

2.8 Main option 8 –R1-R9

R	Input example	Output and fresh-load expectation
R1	Group by city, or choose city Toronto for venue detail Start 2000-01-01, end today;	Original ticket count and face-value revenue by selected location grouping; Toronto contains several venues
R2	Option 1. By segment and genre Option 2. By country Option 3. By country and city Option 4. By country, city and venue	Event and performance counts for the selected rollup
R3	Option 1. Overall ranking Option 2. Ranking by country Option 3. Ranking by city	Organizers ordered by original face-value ticket revenue with stable tie handling
R4	No additional input; choose report option 4	The rolling 12-month report evaluates each customer/city group. Customers 2001 and 2002 qualify in Toronto on a fresh load: each purchased 12 tickets and listed 7; their current restrictions are synchronized
R5	Option 1 takes a date range, e.g., Start 2000-01-01, end today; Option 2 needs no further input	Customer ranking by order count in the period, or one-year rankings within each city

R	Input example	Output and fresh-load expectation
R6	Choose customer cancellations or organizer cancellations; no date input	The fixed preceding-year window includes customers 2086–2097 and organizer-cancelled performances 6058 and 6060
R7	Options 1&2 need no date; Option 3 asks for year and month, e.g., Year: 2026 Month: 7	Sell-through by performance or tier; for the correct generated month, 6003, 6007, 6019 are sold out and 6004, 6022, 6025 are below 25%
R8	Option 1 needs no date; Option 2 takes a date range covering generated resales, e.g., Start 2000-01-01, end today;	Per-event resale statistics and the top 10 events by completed-resale volume; the fresh load contains 12 completed resales
R9	No additional input;	Event IDs 5001–5010, up to 10 phrases

Note R4 has an operational effect: after it runs, a currently qualifying customer is prohibited from booking, listing, and purchasing resale tickets. This is why customers 2001 and 2002 should be used for the R4 demonstration rather than a normal booking demonstration.

2.9 Main option 9 – Organizer toolkit

2.9.1 Recommend pricing

Enter a genre ID, city, and planned venue capacity. A useful sample is

Genre: 4101 (Rock),

City: Toronto,

Capacity: 120.

The system evaluates historical data and generates the following:

- **Recommendation Method:** The search parameters applied (Primary match, Expanded fallback, or Default fallback).
- **Summary Statistics:** Total comparables used, recommended tier count, and average historical primary-ticket revenue.
- **Pricing Tiers Table:** A breakdown of capacity share (totaling 100%) and suggested price per tier.
- **Comparables Log:** A list of influencing past events, including venue details, completion dates, and the specific selection reason (e.g., exact genre vs. same segment).

Search Logic Breakdown:

- **Primary Match:** Past 24 months, target city, exact genre or same segment, +/- 25% capacity.
- **Expanded Fallback:** Past 36 months, any city, +/- 50% capacity.

- **Default Fallback** (if no usable history exists):

Tier	Capacity share	Price
Low	50%	\$60
Middle	30%	\$100
High	20%	\$150

2.9.2 Estimate revenue impact

Enter genre, city, capacity, current price, proposed price, and an optional comparison band.
Example:

Capacity: 120,

Current price: 100,

Proposed price: 120,

Band: 20.

The result reports comparable sample sizes, expected sell-through at each price, expected revenue at each price, and estimated change. This is a historical heuristic, not a guaranteed forecast.

3. System limitations

- The terminal has no authentication or authorization session. An identifier proves which row to use, not who is operating the terminal. In particular, organizer cancellation currently accepts a performance ID without proving that the operator controls its event.
- Payment card values are stored in MySQL. The system does not integrate a payment processor, or chargeback handling.
- Location searches use coordinates, postal prefixes, or normalized exact addresses; they do not call a geocoder and do not account for driving distance.
- Scalper detection follows the required 10-ticket and more-than-half threshold logic, which can produce false positives. A real system would use risk review, appeals, account/device/payment signals, and carefully governed enforcement.
- Toolkit recommendations depend on the quantity and quality of completed historical data. Sparse history triggers a transparent fallback rather than pretending to be precise.
- There is no web/mobile UI, email notification, external payment, horizontal scaling, or production monitoring.

4. Possibilities for improvement

1. Add secure authentication, role-based authorization, and organizer-event ownership checks. Organizer cancellation should at minimum request and verify the organizer ID; a production design should derive it from an authenticated session and require reauthentication for destructive actions.
2. Encrypt sensitive fields, replace stored card numbers with provider tokens, add secret management, and maintain an auditable permission log.
3. Store venue time zones and make local-time display and daylight-saving conversion explicit.
4. Add geocoding and address canonicalization, with spatial indexes for larger datasets.
5. Add a reviewed scalper workflow with explanations, expiration, appeals, and more behavioral signals.
6. Improve toolkit evaluation with confidence intervals, seasonal features, artist popularity, day/time effects, and held-out accuracy measurements.
7. Add automated end-to-end tests for every terminal path and export report results to CSV or PDF.
8. Add a web or mobile interface with searchable selectors so users do not need to know numeric IDs.

5. Understand results and validation

Every operation begins with a status:

Status	Meaning
SUCCESS	The operation finished and committed.
INVALID_INPUT	A value has the wrong format or range. Correct it and retry.
NOT_FOUND	The requested ID or record does not exist.
CONFLICT	Current data makes the action impossible, such as a sold seat or insufficient general admission capacity.
FORBIDDEN	A business rule rejects the action, such as a late cancellation or a restricted customer booking.
DATABASE_ERROR	MySQL rejected the operation or the connection failed.

Within operations, searches, reports, and the organizer toolkit, an invalid value that cannot be used is displayed immediately as **CONFLICT: <reason>**, followed by **Try again? (y/n) : .** Enter **y** to re-enter the affected value or request. Enter **n** to leave that input path without running the invalid request.

6. Sample data IDs

The sample data is regenerated relative to the date on which `sql/load.sql` runs, but these identifier ranges and special cases are stable on a fresh load.

6.1 IDs ranges

Data	IDs	Useful purpose
Organizers	1001-1005	Create events, view managed-event sales, and group organizer reports
Customers	2001-2100	Profiles, histories, booking, resale, reviews, and customer reports
Payment records	2101-2200	Used automatically by purchases
Venues	3001-3008	Event performances, location queries, and pricing
Segments	4001-4003	4001 Music, 4002 Arts, 4003 Sports
Genres	4101-4106	4101 Rock, 4102 Pop, 4103 Musical, 4104 Comedy, 4105 Basketball, 4106 Hockey
Artists/teams	4201-4215	Event billing order
Events	5001-5030	Reports, reviews, performances, and toolkit history
Performances	6001-6070	Inventory, booking, cancellation, Q4-Q7, and reports
Original orders	7001-7320	Customer history and sales reports
Tickets	8001-8980	Cancellation, resale, ownership, and R8

6.2 Best IDs to demonstrate features

IDs/values	What they demonstrate
Organizers 1001-1005	Create events, view managed-event sales, and group organizer reports
Venue 3001-3003, postal M5J 2X2, coordinates 43.643500, -79.379100	Toronto location searches
Exact address 40 Bay Street	Q3 exact-address search
Performance 6001	Mixed reserved/GA inventory, sold and unsold seats, blocked seats, active resale, Q6, and Q7
Performance 6002	Future performance inside the seven-day cancellation boundary
Performances 6003, 6007, 6019	Sold-out cases

IDs/values	What they demonstrate
Organizers 1001-1005	Create events, view managed-event sales, and group organizer reports
Performances 6004, 6022, 6025	Below-25% sell-through cases
Performances 6006, 6013	Future scheduled performances with no ticket history; pricing can be replaced
Performances 6058, 6060	Organizer-cancelled cases
Customers 2001, 2002	R4 possible-scalper examples; booking and resale actions are restricted after R4 synchronization
Customers 2086-2097	Customer cancellation examples for R6
Customer 2011 and performance 6003	An attended performance available for a new review
Ticket 8001	Active reserved ticket with an active listing at its resale cap
Ticket 8002	Active GA ticket with an active listing at its resale cap
Ticket 8004	Future unlisted reserved ticket, suitable for customer cancellation; its owner may be restricted from resale
Ticket 8006	Near-deadline ticket whose customer cancellation should be rejected
Ticket 8007	Ticket with two completed ownership transfers
Tickets 8133-8135, 8139-8141	Completed resale examples
Tickets 8136, 8137, 8142, 8143	Withdrawn listing examples

IDs created by a new operation are printed by the terminal and are not part of the fixed ranges above.